

HARSHITH DUPAM

📍 Hyderabad, Telangana | ✉️ harshithdupam30@gmail.com | 🌐 github.com/HarshDev10218 | 🔗 [linkedin.com/in/harshith-dupam](https://www.linkedin.com/in/harshith-dupam)

EDUCATION

Chaitanya Bharathi Institute of Technology (CBIT) — *Hyderabad, Telangana Bachelor of Technology (B.Tech) in Chemical Engineering* | **CGPA: 10.0 / 10.0** * **Core Academic Focus:** Numerical Analysis, Computational Fluid Dynamics, Computational Methods in Chemical Engineering, Process Modeling & Simulation, Advanced Data Analysis, System Optimization Layouts.

Narayana Junior College — *Hyderabad, Telangana Intermediate Education* — *MPC (Mathematics, Physics, Chemistry)* ---

TECHNICAL SKILLS & COMPETENCIES

- **Programming Languages:** Python, JavaScript (ES6+), C Language, GDScript (Godot Engine), SQL (PostgreSQL)
- **Frontend Web Engineering:** HTML5, CSS3, JavaScript Web APIs, UI/UX Glassmorphic Theme Layouts, Responsive Web Frameworks, Event-Driven UI Architectures
- **Software Development & Systems:** Object-Oriented Programming (OOP), Data Structures & Algorithms (DSA), State Machine Architectures, System Logic Architecture, Real-Time Physics Engines
- **Database & Backend Services:** Supabase Integration, Relational Database Schemas, Asynchronous Data Pipelines, Local File I/O Management
- **Developer Ecosystem Tools:** Git Version Control, GitHub Mainframe, Visual Studio Code (VS Code), Browser DevTools Debugging

TECHNICAL PROJECTS ARCHITECTURE

N//S//P — Neon Survival Protocol (Full-Stack Game & Software Core) *Independent Software Engineer* *
Architected an endless 2D arcade survival simulation application featuring complex rendering loops, engineered entirely over a responsive **React** core engine wrapper.

- Built a high-performance rendering canvas using **HTML5 Canvas** API to handle fluid 60FPS math-based coordinate tracking calculations for active entity instances.
- Integrated a secure, decoupled cloud database layer using **Supabase** client modules to handle asynchronous data synchronization for real-time user validation, player wallet tokens, and live global leaderboard schemas.
- Programmed a custom dynamic Finite State Machine (FSM) AI script that dictates enemy node behavior and sector boss tracking patterns, introducing exponential velocity scaling based on active performance variables.

DSA Visualizer Engine (Desktop Simulation Architecture) *Independent Software Engineer* * Developed an interactive data visualization platform using **Python** designed to step through execution stages of sorting (Bubble, Merge, Quick) and searching (Binary, Linear) algorithms.

- Resolved user interface freezing during high-iteration cycles by breaking recursive algorithm execution pipelines into distinct execution chunks, optimizing thread management.
- Constructed a robust diagnostic benchmark dashboard using **Tkinter** to evaluate, contrast, and visually graph Big-O time and space complexity across varying randomized data load distributions.

Space Defenders Architecture (2D Game System) *Independent Software Engineer* * Designed and compiled a complete space-shooter arcade title using the **Godot Engine** framework and **GDScript**.

- Implemented strict Object-Oriented Programming (OOP) structural design rules to completely decouple the underlying physics calculation engine loop from the interface rendering layers.
- Optimized memory footprints during intense gameplay states by engineering an entity pooling pipeline to systematically reuse and cycle laser/bullet instances instead of triggering heavy instantiation/garbage collection cycles.
- Managed custom vector velocity scripts to handle realistic multi-directional physics, deceleration friction, and smooth viewport boundary clamping protocols.

Retro Snake Simulator (Grid Calculation Engine) *Independent Software Engineer* * Programmed a standalone, customizable classic arcade engine using the **Pygame** framework running on strict coordinate matrix layouts.

- Developed custom self-referential grid checking logic to track structural intersections instantly, eliminating delay lag bugs and ensuring crisp frame collisions.
- Configured local asynchronous JSON profile system file handlers to save, parse, and retain persistent high-score record sheets locally on machine reboots.

KEY ACHIEVEMENTS & CONTRIBUTIONS

- **Top-Tier Academic Standing:** Secured and maintained a perfect **10.0 CGPA framework** throughout core engineering coursework at Chaitanya Bharathi Institute of Technology (CBIT).
- **Open Source Development:** Autonomously built, modularized, and actively managed a structural profile containing **12+ active technical code repositories** on GitHub, showcasing strong version control habits.
- **Analytical Problem Solving:** Consistently applied algorithmic logic, computational methods, and systemic problem-solving frameworks across both software design challenges and chemical process modeling scenarios.